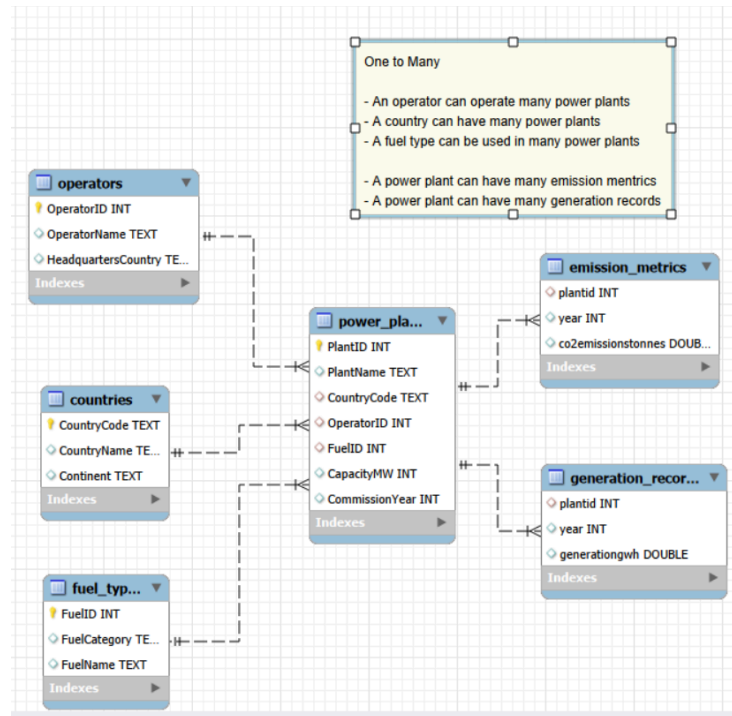


Uche Ogbodo
HW#3

EER Diagram:



1.

```

79
80 -- Part II: SQL Queries
81 -- Q1
82 select
83     power_plants.PlantName,
84     countries.CountryName,
85     operators.OperatorName,
86     fuel_types.FuelCategory,
87     fuel_types.FuelName,
88     power_plants.CapacityMW,
89     power_plants.CommissionYear

```

PlantName	CountryName	OperatorName	FuelCategory	FuelName	CapacityMW	CommissionYear
Three Gorges	China	China Yangtze Power	Renewable	Hydro	22500	2003
Itaipu	Brazil	Electrobras	Renewable	Hydro	14000	1984
Tuoketuo	China	China Yangtze Power	Fossil	Coal	13200	1995
Kashiwazaki-Kariwa	Japan	Tepco Energy	Nuclear	Uranium	8212	1985
Grand Coulee	United States	Global Hydro Inc	Renewable	Hydro	6809	1942
Vindhyachal	India	NTPC Limited	Fossil	Coal	4760	1987
Janschwalde	Germany	Rhein Power AG	Fossil	Coal	3000	1981
Loy Yang	Australia	Snowy Hydro	Fossil	Coal	2210	1984

2.

97

order by power_plants.CapacityMW desc;

98

99

-- Q2

100

select

101

power_plants.PlantName,

102

power_plants.CountryCode,

103

generation_records.year,

104

generation_records.generationwh,

105

emission_metrics.co2emissionstonnes

106

from power_plants

107

join generation_records

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	PlantName	CountryCode	year	generationwh	co2emissionstonnes
▶	Grand Coulee	USA	2024	21500.5	0
	Itaipu	BRA	2024	89000.1	0
	Kashiwazaki-Kariwa	JPN	2024	0	0
	Three Gorges	CHN	2024	98800	0
	Loy Yang	AUS	2024	14200.4	1950000
	Janschwalde	DEU	2024	18400.2	2450000
	Vindhyachal	IND	2024	35600.8	3100000
	Tuoketuo	CHN	2024	71000.3	5200000

3.

115

-- Q3

116

select

117

power_plants.PlantName,

118

power_plants.CountryCode,

119

generation_records.year,

120

generation_records.generationwh,

121

emission_metrics.co2emissionstonnes

122

from power_plants

123

join generation_records

124

on power_plants.PlantID = generation_records.plantID

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	PlantName	CountryCode	year	generationwh	co2emissionstonnes
▶	Grand Coulee	USA	2024	21500.5	0
	Itaipu	BRA	2024	89000.1	0
	Kashiwazaki-Kariwa	JPN	2024	0	0
	Three Gorges	CHN	2024	98800	0
	Loy Yang	AUS	2024	14200.4	1950000
	Janschwalde	DEU	2024	18400.2	2450000
	Vindhyachal	IND	2024	35600.8	3100000
	Tuoketuo	CHN	2024	71000.3	5200000

4.

```

131 -- Q4
132 with OperatorGeneration as (
133     select
134         power_plants.OperatorID,
135         sum(generation_records.generationgwh) as TotalGenerationGWh
136     from power_plants
137     inner join generation_records
138         on power_plants.PlantID = generation_records.plantid
139     group by power_plants.OperatorID

```

OperatorName	HeadquartersCountry	TotalGenerationGWh
China Yangtze Power	CHN	169800.3
Electrobras	BRA	89000.1
NTPC Limited	IND	35600.8
Global Hydro Inc	USA	21500.5
Rhein Power AG	DEU	18400.2
Snowy Hydro	AUS	14200.4
Tepco Energy	JPN	0

5.

```

150 -- Q5
151 with CountryGeneration as (
152     select
153         power_plants.CountryCode,
154         sum(generation_records.generationgwh) as TotalGenerationGWh
155     from power_plants
156     inner join generation_records
157         on power_plants.PlantID = generation_records.plantid
158     group by power_plants.CountryCode
159 ),
160 CountryEmissions as (

```

CountryName	TotalGenerationGWh	TotalCO2EmissionsTonnes
China	169800.3	5200000
Brazil	89000.1	0
India	35600.8	3100000
United States	21500.5	0
Germany	18400.2	2450000
Australia	14200.4	1950000
Japan	0	0